MATH329 · OPERATIONS RESEARCH

Lecture 2 Graphical Method

Midterm summary · Chapter 2

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01 The idea

When a problem has only two variables, x_1 and x_2, we can draw the constraints on paper and see where the best point is. This is a fixed question in every midterm, worth 6 to 9 marks.

The graphical procedure includes two steps:
1. Determination of the feasible solution space.
2. Determination of the optimum solution from among all the feasible points in the solution space.

Term	Meaning
Feasible solution space	The region that satisfies all the constraints together.
Corner point	A corner of the region. The optimum always occurs at a corner.
Optimum	The corner that gives the largest z for max or the smallest z for min.

02 How to draw one constraint

Take the constraint $x_1 + 2x_2 \le 8$ as an example:

1 Make it an equation and find two points

$$x_1 + 2x_2 = 8$$

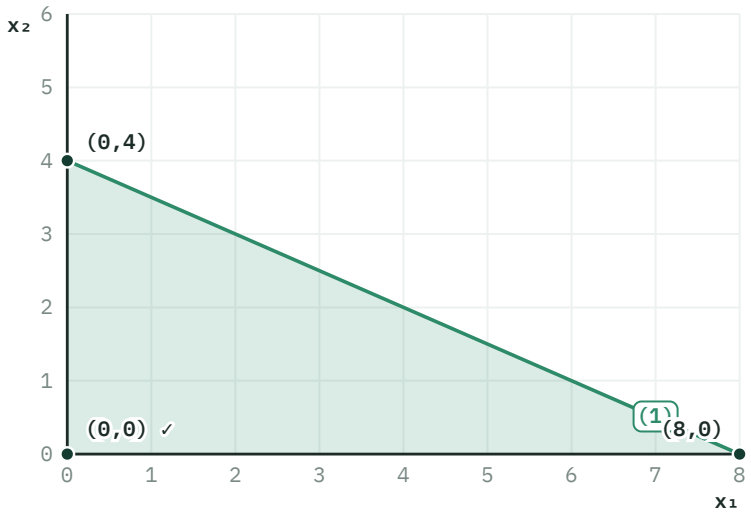
$x_1 = 0 \quad \rightarrow \quad x_2 = 4 \qquad \text{point } (0, 4)$

$x_2 = 0 \quad \rightarrow \quad x_1 = 8 \qquad \text{point } (8, 0)$

2 Pick the side by testing $(0, 0)$

$0 + 2(0) = 0 \le 8 \qquad \text{true} \quad \rightarrow \quad \text{the side that contains } (0, 0)$

If the test is false, the region is on the other side. If the line passes through $(0, 0)$ itself, test another point such as $(1, 0)$.



Form of the constraint	How it is drawn
$x_1 \leq 4$	A vertical line at $x_1 = 4$. No substitution is needed.
$x_2 \leq 3$	A horizontal line at $x_2 = 3$.
$-2x_2 \geq -4$	Multiply by -1 and flip the sign: $2x_2 \leq 4$, that is $x_2 \leq 2$.
$x_1 - x_2 \leq 1$	Its points are $(1, 0)$ and $(0, -1)$. The second point is below the axis, which is normal.
$x_1, x_2 \geq 0$	The region is always in the first quadrant.

03 The full steps

1. Draw every constraint as a line (two points each)
2. Mark the correct side of each line by testing (0, 0)
3. Shade the region that all the constraints share
4. Find every corner point (solve the two lines that meet there)
5. Put each corner in z
6. Choose the largest z (max) or the smallest z (min)
7. Draw the objective line through the optimum and write the answer

The objective line

The answer key gives a mark for a line labelled **Obj**. Draw it this way: put the optimal `z` in the objective function and draw the resulting line. It should touch the region only at the optimal corner.

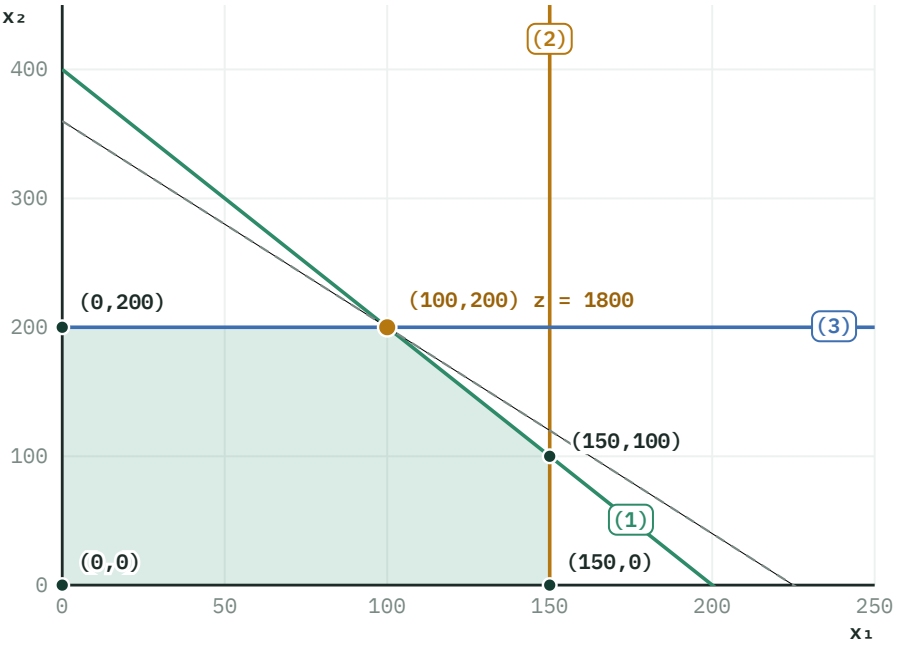
How it is marked

From the answer keys: one mark for each line, one for the objective line, one or two for the solution space, one for the optimum point, and the rest for writing the answer `z`, `x1` and `x2`.

04 Example 2.1 · Wild West (Max)

Maximize $z = 8x_1 + 5x_2$		
subject to	$2x_1 + x_2 \leq 400$	(1)
	$x_1 \leq 150$	(2)
	$x_2 \leq 200$	(3)
	$x_1, x_2 \geq 0$	

The points of line (1) are `(0, 400)` and `(200, 0)`. Line (2) is vertical and line (3) is horizontal.



Corner	Where it comes from	$z = 8x_1 + 5x_2$
(0, 0)	origin	0
(150, 0)	(2) with the x_1 axis	1200
(150, 100)	(1) with (2)	1700
(100, 200)	(1) with (3)	**1800**
(0, 200)	(3) with the x_2 axis	1000

Optimum $z = 1800$ at $x_1 = 100$, $x_2 = 200$

How `(100, 200)` is found: put `x₂ = 200` in (1): `2x₁ + 200 = 400`, so `x₁ = 100`.

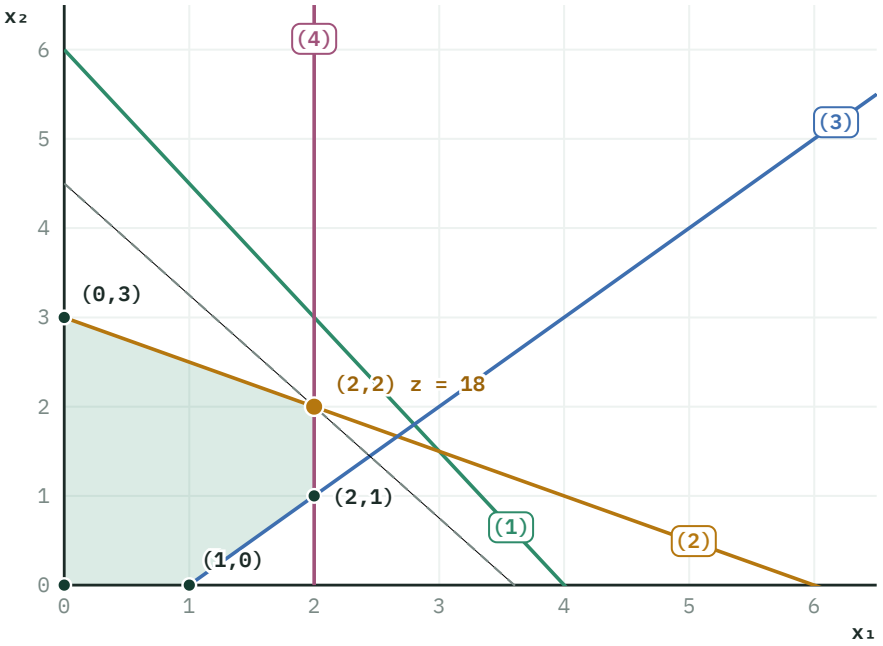
05 Example 2.3 · Reddy Mikks (Max)

Reddy Mikks produces interior and exterior paints from two raw materials, M1 and M2. The daily demand for interior paint cannot exceed that for exterior paint by more than 1 ton, and the maximum daily demand for interior paint is 2 tons.

	Interior (x_1)	Exterior (x_2)	Maximum daily availability
Raw material M1	6	4	24
Raw material M2	1	2	6
Profit per ton ($1000)	5	4	

$$\text{Maximize } z = 5x_1 + 4x_2$$

$$\begin{aligned}
\text{subject to} \quad & 6x_1 + 4x_2 \le 24 && (1) \\
& x_1 + 2x_2 \le 6 && (2) \\
& x_1 - x_2 \le 1 && (3) \quad \text{market limit} \\
& x_1 \le 2 && (4) \quad \text{demand limit} \\
& x_1, x_2 \ge 0
\end{aligned}$$



Corner	Where it comes from	$z = 5x_1 + 4x_2$
(0, 0)	origin	0
(1, 0)	(3) with the x_1 axis	5
(2, 1)	(3) with (4)	14
(2, 2)	(2) with (4)	**18**
(0, 3)	(2) with the x_2 axis	12

Optimum $z = 18$ at $x_1 = 2$, $x_2 = 2$

Notice that line (1) never touches the region. Such a constraint does not affect the solution, but you still draw it because it carries a mark.

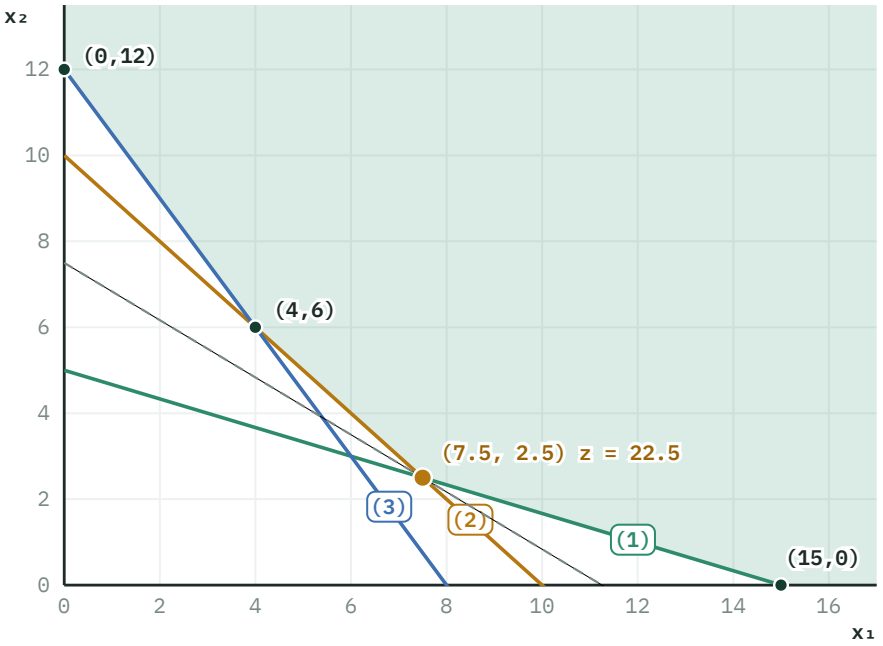
Watch out: the slide graph differs

The graph on the slide comes from the textbook version, whose constraints are $-x_1 + x_2 \le 1$ and $x_2 \le 2$, giving $x_1 = 3$, $x_2 = 1.5$, $z = 21$. The constraints written in the slide text ($x_1 - x_2 \le 1$ and $x_1 \le 2$) give $z = 18$ at $(2, 2)$. In the exam, follow the constraints written in the question.

06 Feed Mix (Min)

Minimize $z = 2x_1 + 3x_2$		
subject to	$x_1 + 3x_2 \ge 15$	(1)
	$2x_1 + 2x_2 \ge 20$	(2)
	$3x_1 + 2x_2 \ge 24$	(3)
	$x_1, x_2 \ge 0$	

All the constraints are $\ge$, so testing $(0, 0)$ is false for all three. The region is on the side away from the origin, and it is open upward and to the right.



Corner	Where it comes from	$z = 2x_1 + 3x_2$
(0, 12)	(3) with the x_2 axis	36
(4, 6)	(2) with (3)	26
(7.5, 2.5)	(1) with (2)	**22.5**
(15, 0)	(1) with the x_1 axis	30

Minimum $z = 22.5$ at $x_1 = 7.5$, $x_2 = 2.5$

How to find where two lines meet

(1) $x_1 + 3x_2 = 15$

(2) $2x_1 + 2x_2 = 20$ → divide by 2: $x_1 + x_2 = 10$

(1) − (2): $2x_2 = 5$ → $x_2 = 2.5$

$x_1 = 10 - 2.5 = 7.5$

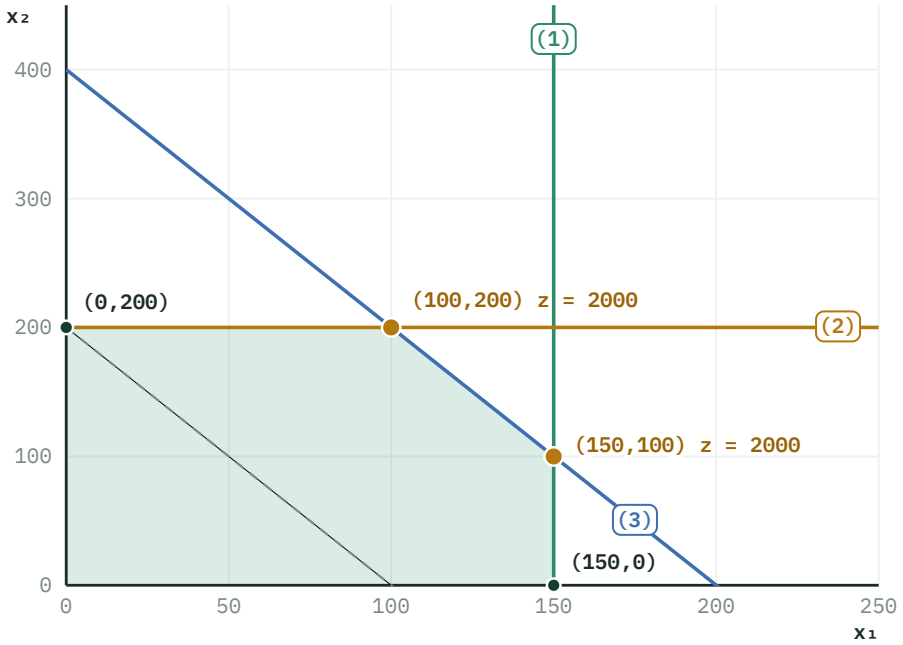
07 Exceptional cases

Usually a problem has one optimal solution. There are three special cases, and they were asked as "mention them" in midterm 462:

1. Alternative optimal solutions
2. Unbounded solution
3. No feasible solution

1 · Alternative optimal solutions

Maximize $z = 10x_1 + 5x_2$		
subject to	$x_1 \le 150$	(1)
	$x_2 \le 200$	(2)
	$2x_1 + x_2 \le 400$	(3)
	$x_1, x_2 \ge 0$	



Two corners give the same value `z = 2000`: `(150, 100)` and `(100, 200)`. Every point on the segment between them gives the same value, so there are infinitely many optimal solutions.

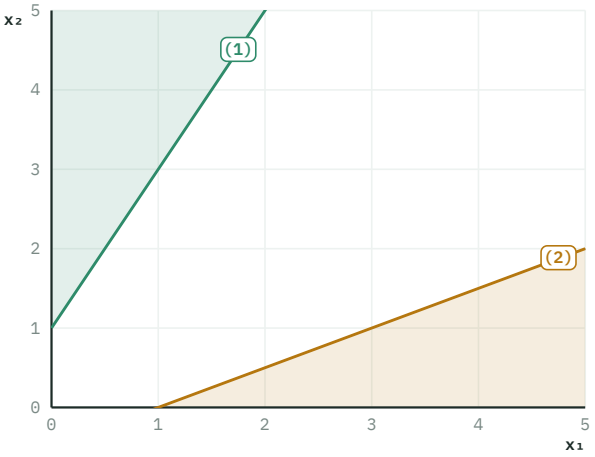
When does it happen?

When the objective function is parallel to one of the constraints. Here `10x₁ + 5x₂` is parallel to `2x₁ + x₂` because the first is five times the second.

2 · No feasible solution

Maximize $z = 5x_1 + 7x_2$
s.t. $2x_1 - x_2 \le -1$ (1)
$-x_1 + 2x_2 \le -1$ (2)
$x_1, x_2 \ge 0$

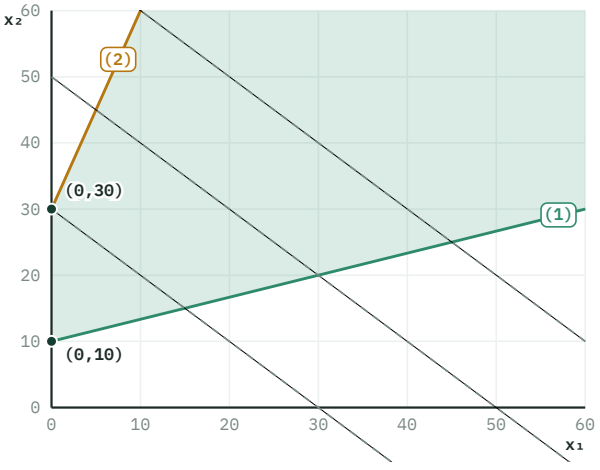
Each constraint has its own region, but the two regions do not overlap in the first quadrant. No point satisfies both constraints.



3 · Unbounded solution

Maximize $z = x_1 + x_2$
s.t. $-x_1 + 3x_2 \ge 30$ (1)
$-3x_1 + x_2 \le 30$ (2)
$x_1, x_2 \ge 0$

The region is open in the direction that increases `z`. The objective line can keep moving up and `z` keeps growing, so there is no finite optimum.



An open region alone does not mean unbounded. The Feed Mix region is open, but it is a min problem and the optimum is at a corner.

08 Common mistakes

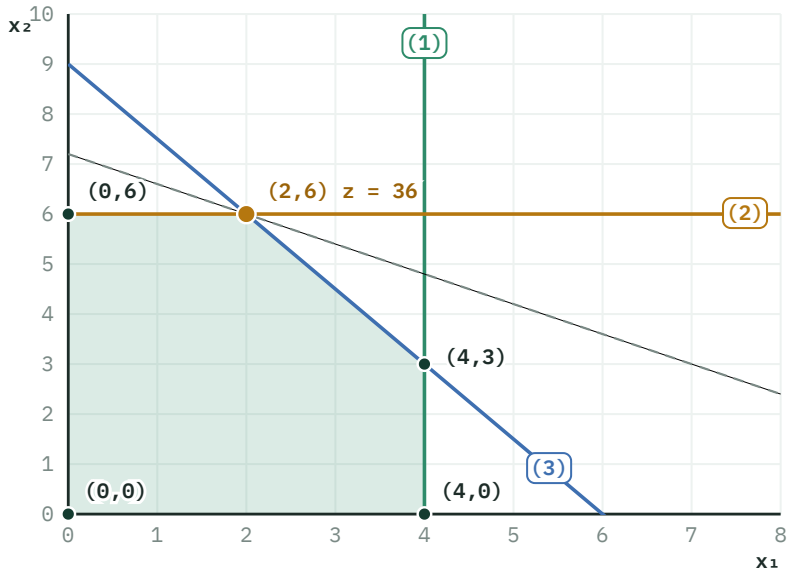
Mistake	Correct
Substituting in a one-variable constraint	$x_1 \le 4$ is simply a vertical line.
Multiplying by -1 without flipping the sign	$-x_1 \ge -4$ becomes $x_1 \le 4$.
Reading a corner off the graph by eye	Solve the two equations. The answer can be a fraction such as $(4.5, 1)$.
Using a point outside the region	Check that the point satisfies every constraint before putting it in z.
Forgetting the objective line	It carries a mark in the answer key.
Writing z without x_1 and x_2	$Z = 7.5$ at $x_1 = 0.5$ and $x_2 = 3$

09 Test yourself

1. What are the two steps of the graphical procedure?
2. List the three special cases of the graphical solution.
3. When do alternative optimal solutions occur?
4. Solve graphically: $\text{Max } z = 3x_1 + 5x_2$ subject to $x_1 \leq 4$, $2x_2 \leq 12$ and $3x_1 + 2x_2 \leq 18$.

Answers

#	Answer
1	Determination of the feasible solution space, then determination of the optimum solution.
2	Alternative optimal solutions, unbounded solution, no feasible solution.
3	When the objective function is parallel to one of the constraints.
4	Corners: $(0,0)$, $(4,0)$, $(4,3)$, $(2,6)$, $(0,6)$, with z = 0, 12, 27, 36, 30. The answer is $z = 36$ at $(2, 6)$.


x₂
10
9
8
7
6
5
4
3
2
1
0
0
1
2
3
4
5
6
7
8
x₁
(1)
(2)
(3)
(0,6)
(2,6) z = 36
(4,3)
(0,0)
(4,0)